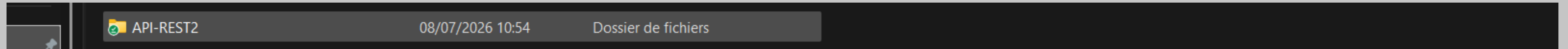


Projet Développement d'API-REST



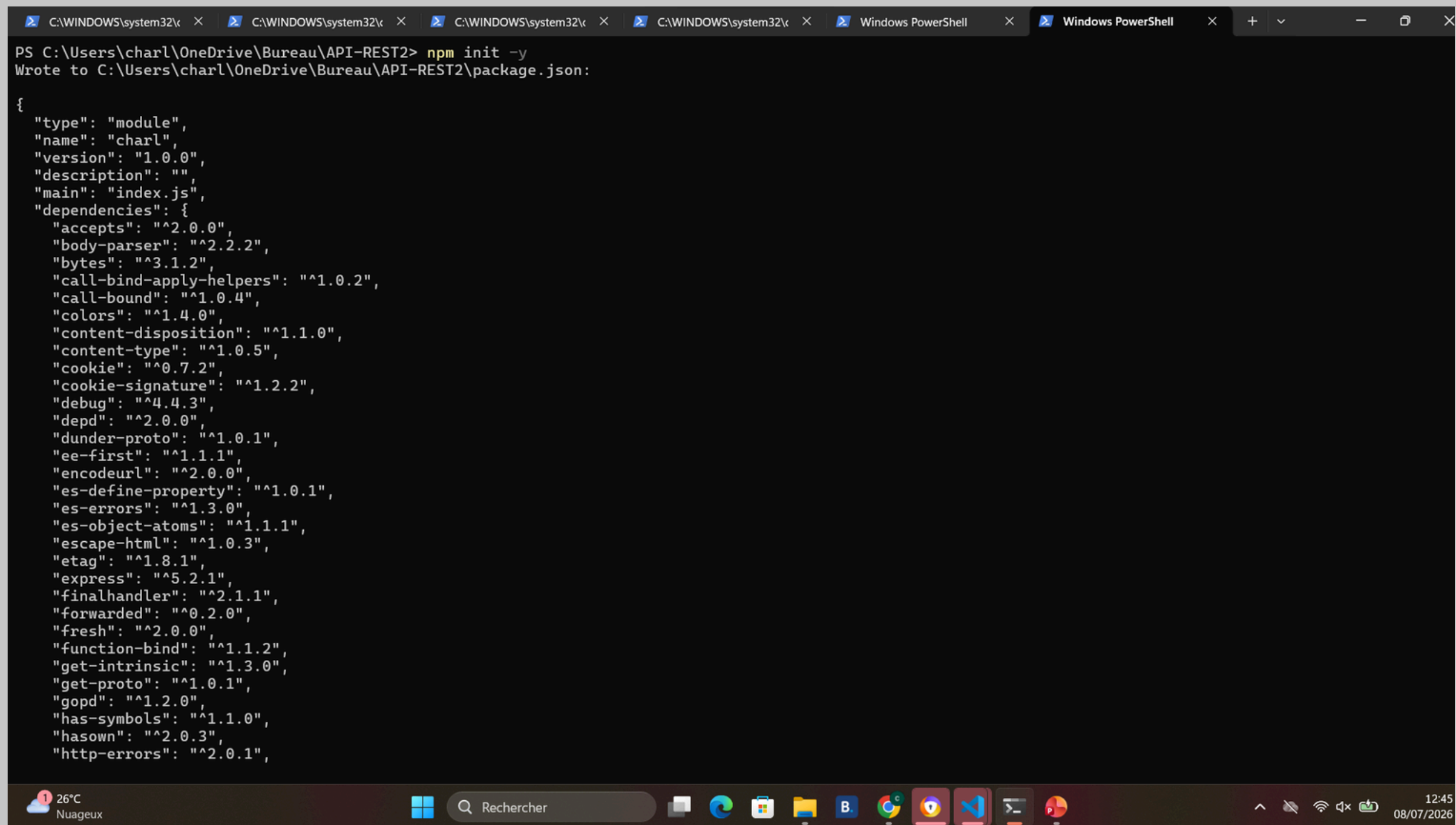
Création d'un nouveau répertoire dans le projet



Différentes commandes utilisés:

Pour le lancement de la console et l'initialisation du projet:

> npm init -y



```
PS C:\Users\charl\OneDrive\Bureau\API-REST2> npm init -y
Wrote to C:\Users\charl\OneDrive\Bureau\API-REST2\package.json:

{
  "type": "module",
  "name": "charl",
  "version": "1.0.0",
  "description": "",
  "main": "index.js",
  "dependencies": {
    "accepts": "^2.0.0",
    "body-parser": "^2.2.2",
    "bytes": "^3.1.2",
    "call-bind-apply-helpers": "^1.0.2",
    "call-bound": "^1.0.4",
    "colors": "^1.4.0",
    "content-disposition": "^1.1.0",
    "content-type": "^1.0.5",
    "cookie": "^0.7.2",
    "cookie-signature": "^1.2.2",
    "debug": "^4.4.3",
    "depd": "^2.0.0",
    "dunder-proto": "^1.0.1",
    "ee-first": "^1.1.1",
    "encodeurl": "^2.0.0",
    "es-define-property": "^1.0.1",
    "es-errors": "^1.3.0",
    "es-object-atoms": "^1.1.1",
    "escape-html": "^1.0.3",
    "etag": "^1.8.1",
    "express": "^5.2.1",
    "finalhandler": "^2.1.1",
    "forwarded": "^0.2.0",
    "fresh": "^2.0.0",
    "function-bind": "^1.1.2",
    "get-intrinsic": "^1.3.0",
    "get-proto": "^1.0.1",
    "gopd": "^1.2.0",
    "has-symbols": "^1.1.0",
    "hasown": "^2.0.3",
    "http-errors": "^2.0.1",
```


Lancement de la console:

> npm install express

```
PS C:\Users\charl\OneDrive\Bureau\API-REST2> npm install express
```

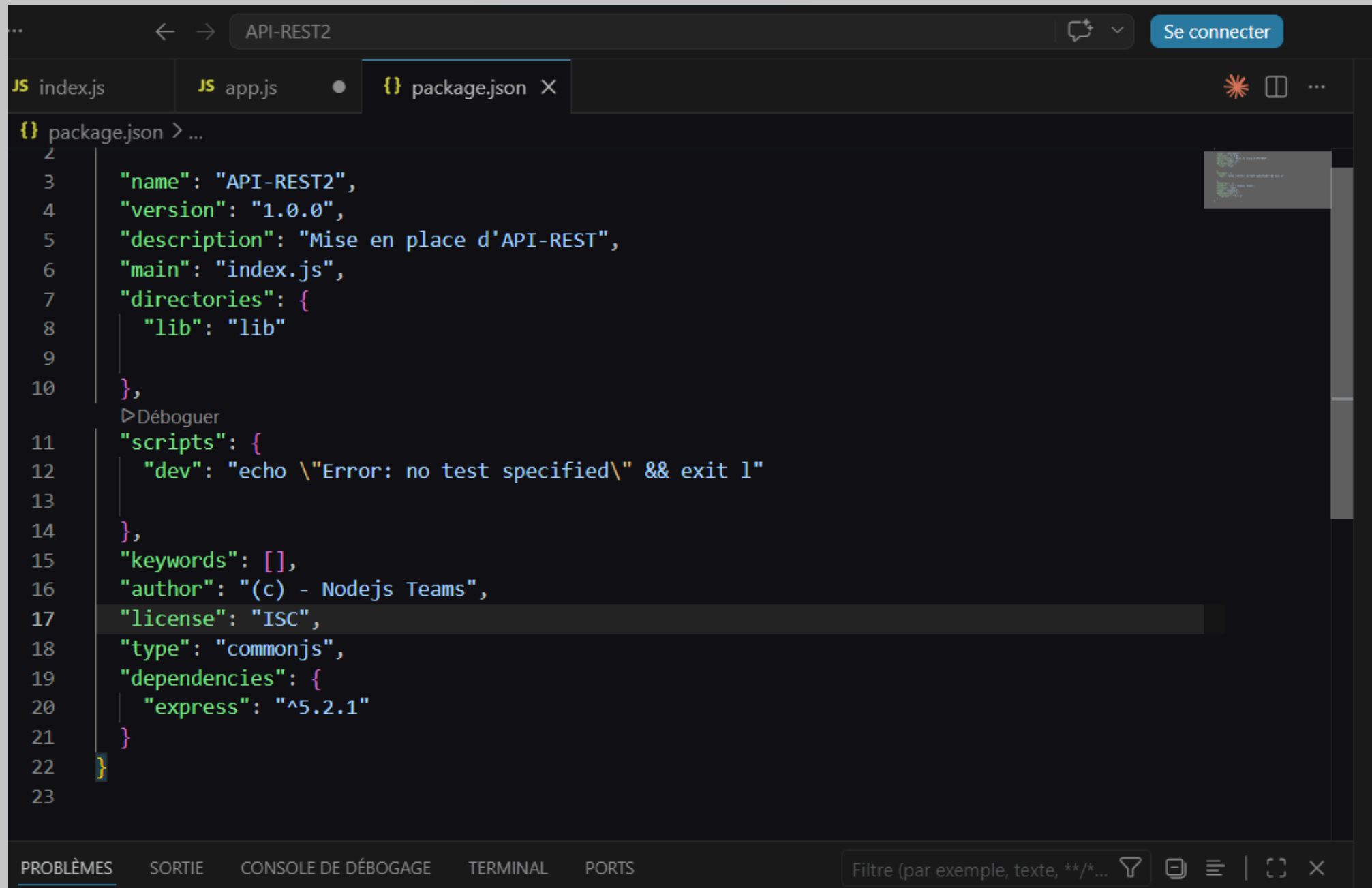
```
up to date, audited 69 packages in 2s
```

```
25 packages are looking for funding  
  run `npm fund` for details
```

```
found 0 vulnerabilities
```

```
PS C:\Users\charl\OneDrive\Bureau\API-REST2> |
```

Pesonnalisaion du package.json généré à l'initialisation



```
2
3  "name": "API-REST2",
4  "version": "1.0.0",
5  "description": "Mise en place d'API-REST",
6  "main": "index.js",
7  "directories": {
8    "lib": "lib"
9  },
10
11  >Déboguer
12  "scripts": {
13    "dev": "echo \"Error: no test specified\" && exit 1"
14  },
15  "keywords": [],
16  "author": "(c) - Nodejs Teams",
17  "license": "ISC",
18  "type": "commonjs",
19  "dependencies": {
20    "express": "^5.2.1"
21  }
22 }
23
```

Lancement du serveur

: Création d'un fichier app.js

```
import colors from 'colors';  
//  
import express from 'express';  
  
const app = express();  
const port = 3003;  
const host = 'localhost';  
  
/*  
*-----
```

```
*      GET
*  -----
*/

app.get('/', (req, res) => {
    res.send('Home page');
});

/*-----
*/
/*      ADMIN DESKTOP
*/
/*-----
*/

app.listen( port, () => {
    console.log( );

    console.log('-----
    -----');
    console.log('  Node.js web server framework running at
http://${host}:{port}  ');

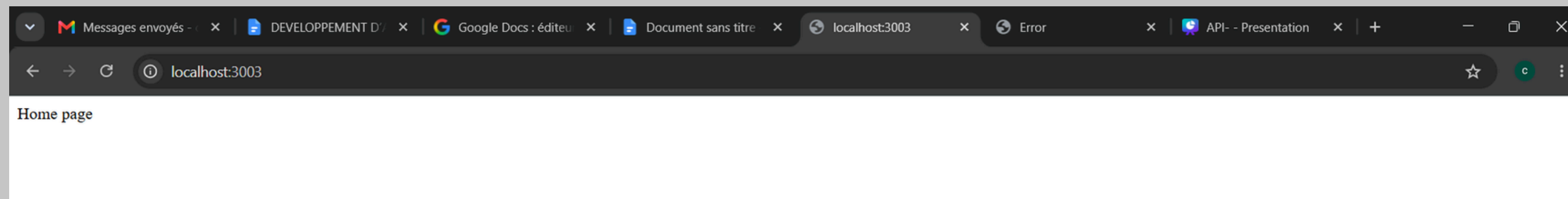
    console.log('-----
    -----');

});
```

Résultat de la commande: > node app.js

```
S C:\Users\charl\OneDrive\Bureau\API-REST2> node app.js

-----
Node.js web server framework running at http://${host}:{port}
-----
S C:\Users\charl\OneDrive\Bureau\API-REST2> |
```



×

Headers

Preview

Response

Initiator

>>

▼ General

Request URL

http://localhost:3003/test

Request Method

GET

Status Code

●

304 Not Modified

Remote Address

[::1]:3003

Referrer Policy

strict-origin-when-cross-origi
n

▼ Response headers

☐ Raw

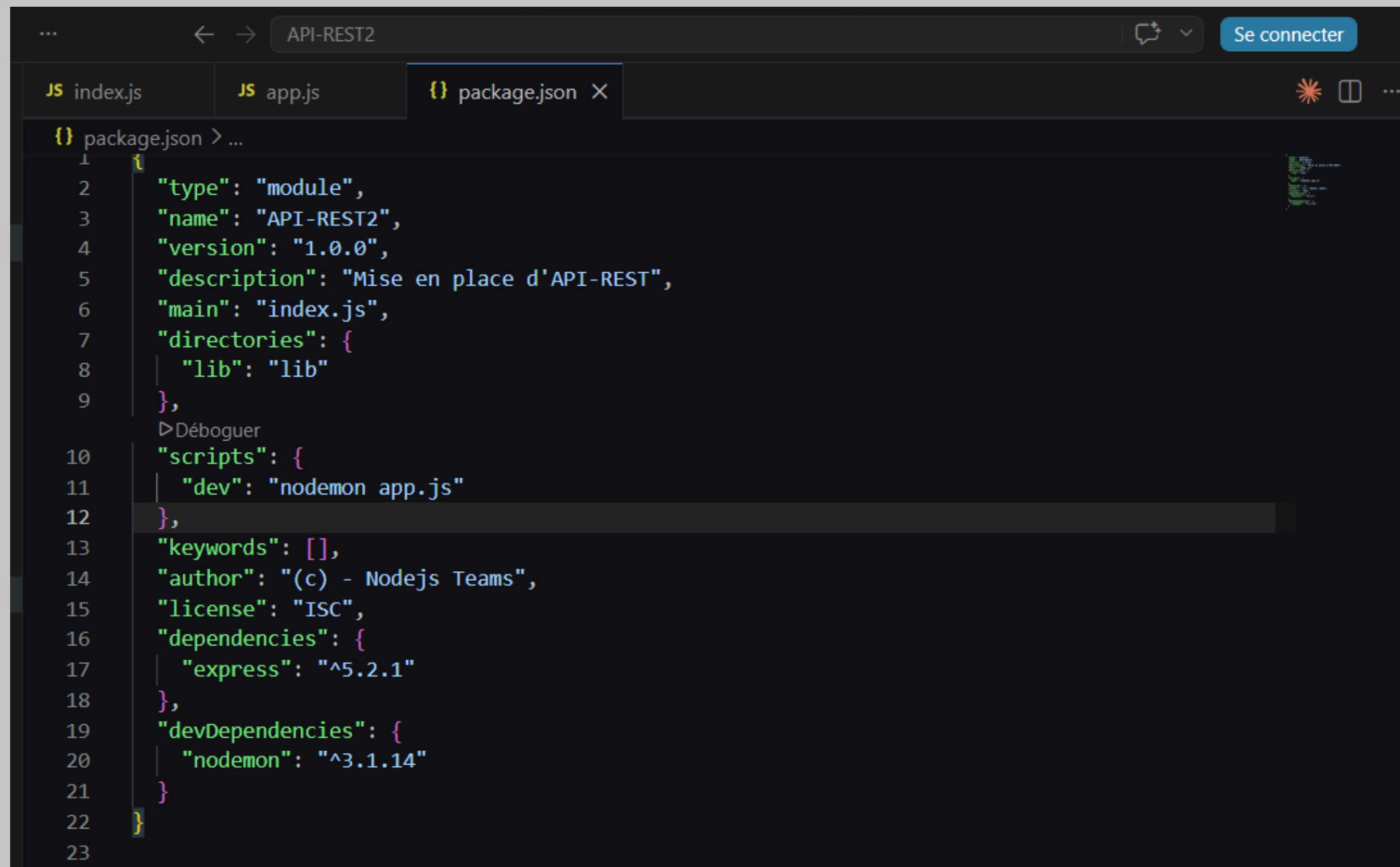
6

Pour éviter de redémarrer le serveur à chaque instant il faut entrer la commande:

> npm i nodemon --save-dev

```
PS C:\Users\charl\OneDrive\Bureau\API-REST2> npm i nodemon --save-dev
added 26 packages, removed 1 package, and audited 94 packages in 5s
30 packages are looking for funding
  run `npm fund` for details
found 0 vulnerabilities
PS C:\Users\charl\OneDrive\Bureau\API-REST2>
```

Modification du package.json en mettant: > “dev”: “nodemon app.js”



The screenshot shows a code editor with three tabs: `index.js`, `app.js`, and `package.json`. The `package.json` tab is active, displaying the following JSON content:

```
1 {  
2   "type": "module",  
3   "name": "API-REST2",  
4   "version": "1.0.0",  
5   "description": "Mise en place d'API-REST",  
6   "main": "index.js",  
7   "directories": {  
8     "lib": "lib"  
9   },  
10  "scripts": {  
11    "dev": "nodemon app.js"  
12  },  
13  "keywords": [],  
14  "author": "(c) - Nodejs Teams",  
15  "license": "ISC",  
16  "dependencies": {  
17    "express": "^5.2.1"  
18  },  
19  "devDependencies": {  
20    "nodemon": "^3.1.14"  
21  }  
22 }  
23
```

The editor interface includes a top bar with navigation arrows, the file name "API-REST2", a search icon, and a "Se connecter" button. The left sidebar shows the file explorer with the current file structure.

Résultat de la commande

> npm run dev

```
27 /
28 GET /entities

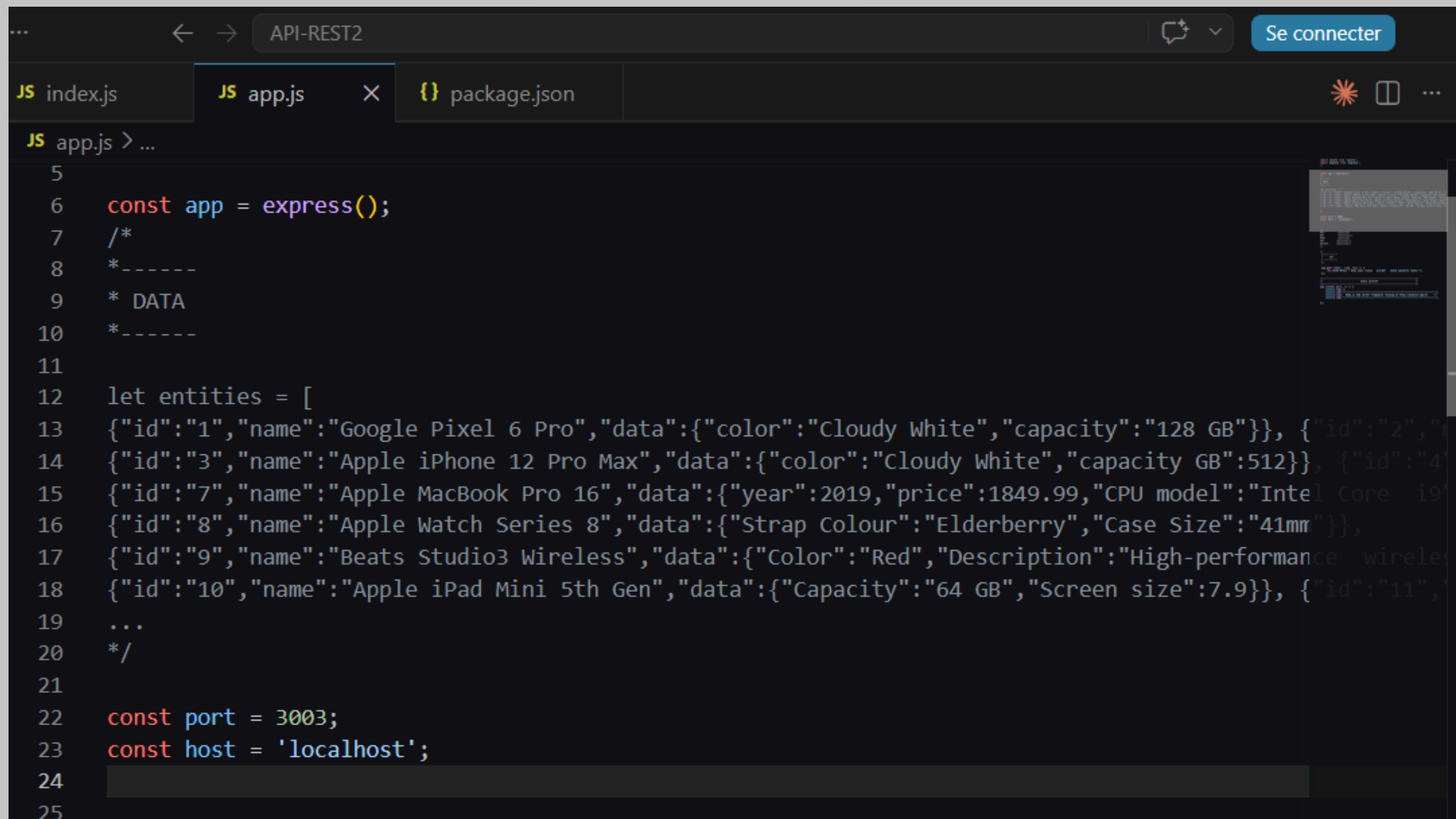
PROBLÈMES  SORTIE  CONSOLE DE DÉBOGAGE  TERMINAL  PORTS

PS C:\Users\charl\OneDrive\Bureau\API-REST2> npm run dev
[nodemon] 3.1.14
[nodemon] to restart at any time, enter `rs`
[nodemon] watching path(s): *.*
[nodemon] watching extensions: js,mjs,cjs,json
[nodemon] starting `node app.js`

-----
Node.js web server framework running at http://${host}:{port}
-----
[nodemon] clean exit - waiting for changes before restart
```

Mise en Place de la liste des éléments de l'API-REST

Ajouts d'éléments dans le app.js après la déclaration de app



```
...  API-REST2  Se connecter

JS index.js  JS app.js  {} package.json

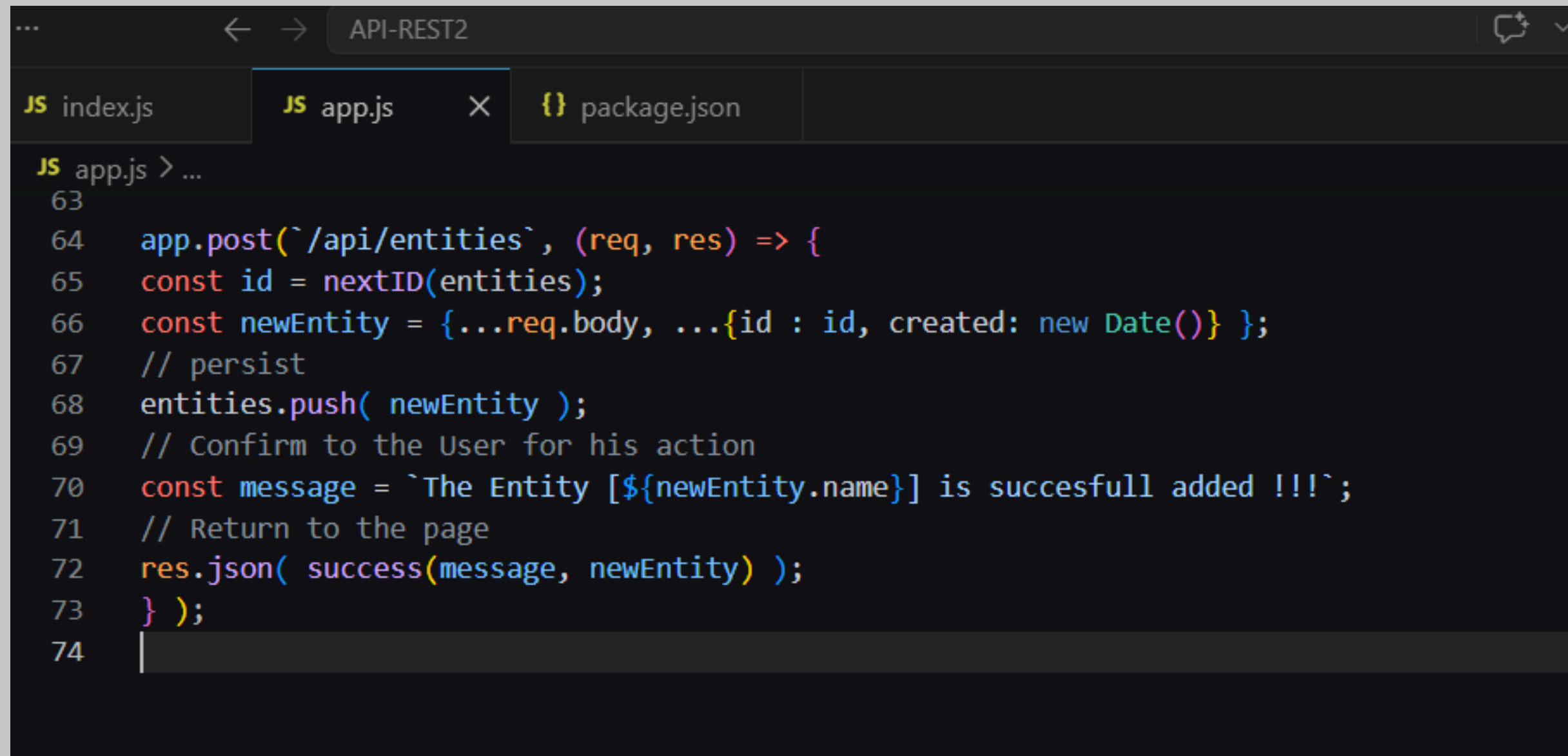
JS app.js > ...
5
6  const app = express();
7  /*
8  * -----
9  *  DATA
10 * -----
11
12  let entities = [
13  {"id":"1","name":"Google Pixel 6 Pro","data":{"color":"Cloudy White","capacity":"128 GB"}}, {"id":"2","
14  {"id":"3","name":"Apple iPhone 12 Pro Max","data":{"color":"Cloudy White","capacity GB":512}}, {"id":"4
15  {"id":"7","name":"Apple MacBook Pro 16","data":{"year":2019,"price":1849.99,"CPU model":"Intel Core i9
16  {"id":"8","name":"Apple Watch Series 8","data":{"Strap Colour":"Elderberry","Case Size":"41mm"}},
17  {"id":"9","name":"Beats Studio3 Wireless","data":{"Color":"Red","Description":"High-performance wireles
18  {"id":"10","name":"Apple iPad Mini 5th Gen","data":{"Capacity":"64 GB","Screen size":7.9}}, {"id":"11",
19  ...
20  */
21
22  const port = 3003;
23  const host = 'localhost';
24
25
```



```
JS index.js JS app.js X {} package.json
JS app.js > ...
41
42 app.get(`/api/entities`, (req, res) => {
43   const message = `List of all entities : (${entities.length})`;
44   res.json( success(message, entities) );
45 });
46
47 app.get(`/api/entities/:id`, (req, res) => {
48   const entity = entities.find( entity => entity.id === parseInt(req.params.id) );
49   const message = "We found one Entity !!!";
50   res.json( success(message, entity) );
51 });
52
```

Utilisation de la commande : “GET” pour implémenter le serveur

Utilisation de la commande “POST” pour ajouter un nouvel élément à la collection



The image shows a code editor interface with a dark theme. At the top, there is a tab labeled "API-REST2" with navigation arrows and a refresh icon. Below the tabs, there are three open files: "index.js", "app.js", and "package.json". The "app.js" file is currently selected and displays the following JavaScript code:

```
JS app.js > ...
63
64 app.post(`/api/entities`, (req, res) => {
65   const id = nextID(entities);
66   const newEntity = {...req.body, ...{id : id, created: new Date()}} };
67   // persist
68   entities.push( newEntity );
69   // Confirm to the User for his action
70   const message = `The Entity [${newEntity.name}] is succesfull added !!!`;
71   // Return to the page
72   res.json( success(message, newEntity) );
73 } );
74
```

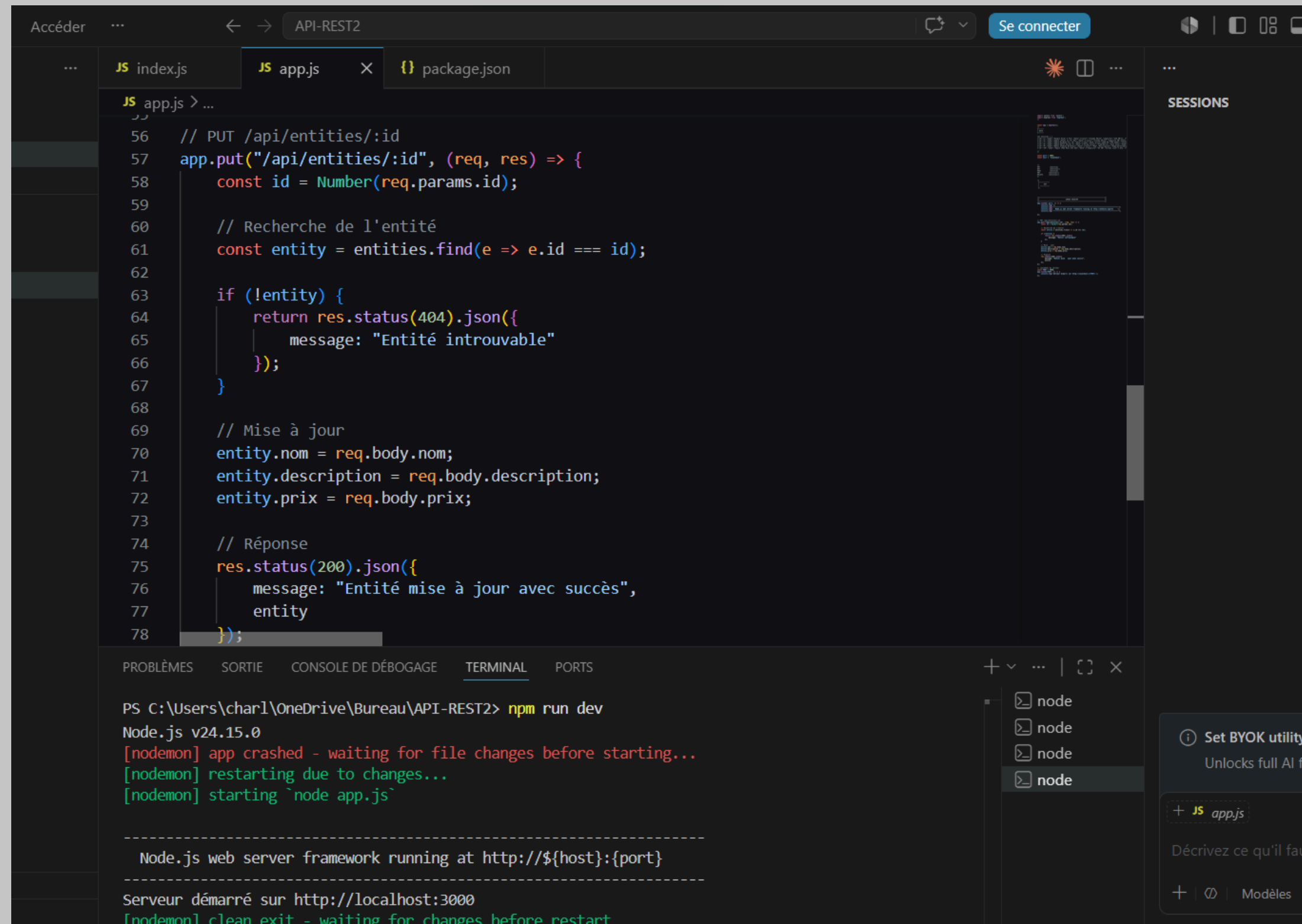
Résultat

```
PROBLÈMES  SORTIE  CONSOLE DE DÉBOGAGE  TERMINAL  PORTS

PS C:\Users\charl\OneDrive\Bureau\API-REST2> npm run dev
Node.js web server framework running at http://${host}:{port}
-----
[nodemon] clean exit - waiting for changes before restart
[nodemon] restarting due to changes...
[nodemon] starting `node app.js`

-----
Node.js web server framework running at http://${host}:{port}
-----
[nodemon] clean exit - waiting for changes before restart
█
```

Proposition de code avec “PUT” pour implémenter la modification de l’algorithme



```
Accéder ... API-REST2 Se connecter
```

```
JS index.js JS app.js X {} package.json
```

```
JS app.js > ...
56 // PUT /api/entities/:id
57 app.put("/api/entities/:id", (req, res) => {
58   const id = Number(req.params.id);
59
60   // Recherche de l'entité
61   const entity = entities.find(e => e.id === id);
62
63   if (!entity) {
64     return res.status(404).json({
65       message: "Entité introuvable"
66     });
67   }
68
69   // Mise à jour
70   entity.nom = req.body.nom;
71   entity.description = req.body.description;
72   entity.prix = req.body.prix;
73
74   // Réponse
75   res.status(200).json({
76     message: "Entité mise à jour avec succès",
77     entity
78   });
79 }
```

```
PROBLÈMES SORTIE CONSOLE DE DÉBOGAGE TERMINAL PORTS
```

```
PS C:\Users\charl\OneDrive\Bureau\API-REST2> npm run dev
Node.js v24.15.0
[nodemon] app crashed - waiting for file changes before starting...
[nodemon] restarting due to changes...
[nodemon] starting `node app.js`

-----
Node.js web server framework running at http://${host}:{port}
-----

Serveur démarré sur http://localhost:3000
[nodemon] clean exit - waiting for changes before restart
```

```
node
node
node
node
```

```
Set BYOK utility m
Unlocks full AI feat

+ JS app.js

Décrivez ce qu'il faut d

+ | | Modèles |
```